

Table 1: Residual-objective ablation under the Double Sphere model. All modes share initialization, splits, and frozen holdout observations. Spherical provides the best macro-average accuracy and cross-sequence ray consistency, while Hybrid is competitive in the central field of view.

<i>(a) Held-out reprojection RMSE [px] ↓</i>				
Residual	Seq. A	Seq. B	Seq. C	Mean
Pixel	0.54014	0.47778	0.54252	0.52015
Spherical	0.53999	0.47710	0.54133	0.51947
Hybrid	0.53995	0.47746	0.54148	0.51963
Kalibr (DS)	0.66693	0.54995	0.63417	0.61702
<i>(b) View-angle-conditioned ray accuracy</i>				
Residual	0–30°	30–50°	≥ 50°	Overall
	True angular RMSE [mrad] ↓			
Pixel	0.3425	0.3423	0.3991	0.3512
Spherical	0.3421	0.3422	0.3963	0.3506
Hybrid	0.3423	0.3421	0.3976	0.3507
<i>(c) Accuracy, geometric stability, and efficiency</i>				
Residual	Overall	Outer	Cross-seq.	Time
	RMSE [px]	RMSE [px]	Ray RMS [deg]	[s]
Pixel	0.5201	0.1249	0.7616	8.5
Spherical	0.5195	0.1262	0.7349	38.3
Hybrid	0.5196	0.1261	0.7477	41.0

Protocol. Seq. A/B/C denote 1444190-clear, 134853-clear, and 192347-clear. All experiments use a 70/30 split with seed 4242. Polar bins are assigned using the shared initialization camera. Cross-sequence ray RMS measures pairwise disagreement over 1,422 shared image locations. Best results are in bold.